

Hedge Funds Crowding Monitor — US

22 May 2026



Kartik Singhal
Goldman Sachs International
Global Banking & Markets



Tanya Ratra
Goldman Sachs & Co. LLC
Global Banking & Markets



Marco Laicini
Goldman Sachs International
Global Banking & Markets

With AI and momentum top of mind, we observe the following notable movements in crowding across US sectors and themes this week:

- **Capital Goods (SPLRCIC)** is the most long crowded sector, with the long crowding score rising from 0.31 to 0.57, now sitting in the 98th percentile over the trailing six-month period (see Exhibits 2 and 4).
- **Software & Services (SPLRCIS)** saw a large decline in short crowding over the past month, with the short crowding score falling to 0.92 from 1.10, now sitting in the 51st percentile versus the trailing six-month period (see Exhibits 2 and 5).
- Amidst geopolitical uncertainty, **Energy (SPNY)** long crowding fell sharply to 0.03 from 0.47, while the short side flipped from uncrowded to crowded, rising to 0.32 from -0.17, suggesting hedge funds have rotated out of long energy positions and begun building shorts (see Exhibit 2).
- **Broad AI (GSTMTAIP)** long crowding continued to build, rising from 0.62 to 0.80, its highest level over the trailing six months (99th percentile, see Exhibits 3 and 6). Meanwhile, **AI-at-Risk (GSTMTAIR)** has become more short crowded, score rising from 0.79 to 0.89 in the last month.
- **AI Data Centers (GSTMTDAT)** long crowding increased to 0.86 from 0.60, now sitting at the 99th percentile over the trailing six-month period, while shorts have been closed with the score declining to -0.42 from -0.21 (see Exhibits 3 and 6).
- **High Momentum Winners (GSXUHMOM)** long crowding has intensified meaningfully, with the score rising to 0.93 from 0.69 over the past month (99th percentile vs past 6m). On the short side, crowding has eased further to -0.28 from -0.16. (see Exhibits 3 and 6).

Crowding scores are expressed as z-scores — measuring the number of standard deviations from the mean — and typically range from -2 to +2. A score above 0 indicates crowded positioning, while a score above +1 indicates highly crowded positioning. For full methodology and definitions, refer to the Methodology and Definitions section below.

Crowding Monitor

Exhibit 1: Crowding Overview: US Sectors and Themes

Ticker	Name	Long Crowding Score				Short Crowding Score			
		Spot	-1m	Change	6m %-ile	Spot	-1m	Change	6m %-ile
SPXBK	Banks	(0.03)	(0.27)	slightly more crowded now	21%	(1.00)	(1.01)	uncrowded but less	51%
SPLRCIC	Capital Goods	0.57	0.31	now even more crowded	98%	0.37	0.39	crowded but less	92%
SPLRCCS	Comm. & Prof. Svcs.	(0.96)	(1.12)	uncrowded but less	83%	0.92	0.89	now even more crowded	91%
SPLRCFD	Cons. Stap. Distribution & Retail	(0.67)	(0.31)	now even more uncrowded	28%	1.31	0.97	now even more crowded	90%
SPLRCLP	Consumer Durables & Apparel	(0.40)	(0.56)	uncrowded but less	69%	0.69	0.65	now even more crowded	21%
SPLRCHR	Consumer Services	0.13	0.21	crowded but less	66%	1.14	1.03	now even more crowded	78%
SPNY	Energy	0.03	0.47	slightly less crowded now	45%	0.32	(0.17)	crowded now from uncrowded	71%
SPLRCREC	Equity REITs	(0.99)	(1.30)	uncrowded but less	93%	(0.22)	0.07	uncrowded now	1%
SPLRCDF	Financial Services	(0.13)	0.16	uncrowded now from crowded	35%	(0.23)	(0.47)	uncrowded but less	78%
SPLRCFBT	Food, Beverage & Tobacco	(0.99)	(1.03)	uncrowded but less	10%	0.39	0.46	crowded but less	82%
SPLRCHC	Health Care Equipment & Services	(0.02)	0.01	no change	60%	(0.20)	(0.31)	uncrowded but less	95%
SPLRCHPE	Household & Personal Products	(0.06)	(0.45)	slightly more crowded now	69%	0.80	0.77	now even more crowded	74%
SPXIN	Insurance	(1.54)	(1.65)	uncrowded but less	6%	(0.17)	(0.22)	uncrowded but less	99%
SPLRCMA	Materials	0.13	0.03	crowded now	34%	(0.09)	(0.07)	no change	81%
SPLRCME	Media & Entertainment	0.14	0.28	crowded but less	10%	(1.54)	(1.06)	now even more uncrowded	2%
SPLRCPHB	Pharma, Biotech & Life Sciences	(0.17)	(0.16)	now even more uncrowded	10%	(0.33)	(0.01)	uncrowded now	17%
SPLRCSE	Semi & Semi Equipment	0.32	0.30	now even more crowded	62%	0.16	(0.23)	crowded now from uncrowded	52%
SPLRCIS	Software & Services	0.05	0.15	slightly less crowded now	39%	0.92	1.10	crowded but less	51%
SPLRCTEHW	Technology Hardware & Equipment	0.14	0.18	crowded but less	47%	0.62	0.65	crowded but less	19%
SPLRCTRN	Transportation	0.39	0.49	crowded but less	2%	1.03	0.96	now even more crowded	77%
SPLRCU	Utilities	(0.72)	(0.82)	uncrowded but less	15%	0.19	0.09	crowded now	70%

Ticker	Name	Long Crowding Score				Short Crowding Score			
		Spot	-1m	Change	6m %-ile	Spot	-1m	Change	6m %-ile
GSTMTAIR*	AI At Risk	0.15	0.29	crowded but less	22%	0.89	0.79	now even more crowded	81%
GSTMTDAT*	AI Data Centers	0.86	0.60	now even more crowded	99%	(0.42)	(0.21)	now even more uncrowded	0%
GSX1BIOT*	Biotech (Index)	0.01	(0.11)	slightly more crowded now	64%	(0.25)	(0.28)	uncrowded but less	78%
GSCBBTC1*	Bitcoin Sensitive Eq	0.82	0.67	now even more crowded	90%	0.78	0.93	crowded but less	6%
GSTMTAIP*	Broad AI	0.80	0.62	now even more crowded	99%	(0.15)	0.03	uncrowded now	0%
GSXUCHSE*	China Sales Exposure	0.56	0.35	now even more crowded	98%	0.06	0.10	no change	0%
GSX1CYCL*	Cyclicals ex Commod	(0.15)	(0.19)	uncrowded but less	52%	0.38	0.30	now even more crowded	95%
GSXUDFNS*	Defense	0.02	(0.22)	slightly more crowded now	99%	(0.32)	(0.35)	uncrowded but less	73%
GSX1DEFS*	Defensives ex Commod	(0.64)	(0.76)	uncrowded but less	66%	0.19	0.16	now even more crowded	83%
GSX1SF8X*	Expensive Software (Index)	0.01	(0.11)	slightly more crowded now	79%	0.77	0.80	crowded but less	59%
GVIPTR	HF VIP Index	1.38	1.28	now even more crowded	75%	(0.13)	(0.05)	uncrowded now	0%
GSXUHMOM	High Momentum (Winners)	0.93	0.69	now even more crowded	99%	(0.28)	(0.16)	now even more uncrowded	9%
GSCBRGBK*	Liquid Regional Banks	(0.49)	(0.59)	uncrowded but less	96%	(0.81)	(0.79)	now even more uncrowded	36%
GSXULOWQ	Low Quality	0.17	0.06	crowded now	95%	0.23	0.20	now even more crowded	62%
GSX1MEGA*	Mag7	(0.04)	0.05	no change	22%	(0.11)	(0.06)	uncrowded now	35%
GSXUMEME*	Memes	(0.08)	(0.07)	no change	44%	1.20	1.28	crowded but less	0%
GSXUMETL*	Metals	0.48	0.13	now even more crowded	98%	(0.46)	(0.50)	uncrowded but less	79%
GSXENATG*	Nat Gas	0.64	0.77	crowded but less	10%	(0.04)	(0.16)	slightly more crowded now	76%
GSX1NPTC*	Non-Profitable Tech	0.18	(0.12)	crowded now from uncrowded	99%	0.59	0.58	now even more crowded	22%
GSXURANI*	Nuclear	(0.11)	(0.12)	uncrowded but less	52%	0.24	0.20	now even more crowded	91%
GSXLCBNI*	Obesity Drugs	0.12	0.20	crowded but less	83%	0.30	0.62	crowded but less	23%
GSX1POW1*	Power Up America	0.76	0.60	now even more crowded	40%	0.17	0.27	crowded but less	46%
GSXURNEW*	Renewables	0.30	(0.10)	crowded now	97%	0.17	0.27	crowded but less	23%
GSXURFAV*	Retail Favorites	0.83	0.92	crowded but less	28%	0.71	0.76	crowded but less	3%
GSCBMSAL	Rolling Most Short	0.09	(0.14)	slightly more crowded now	98%	0.52	0.42	now even more crowded	67%

*The composition and weights of these thematic baskets reflect the latest available data and are held constant across all historical periods to ensure consistent comparison over time

Exhibit 2: Sector Crowding: Long & Short Side

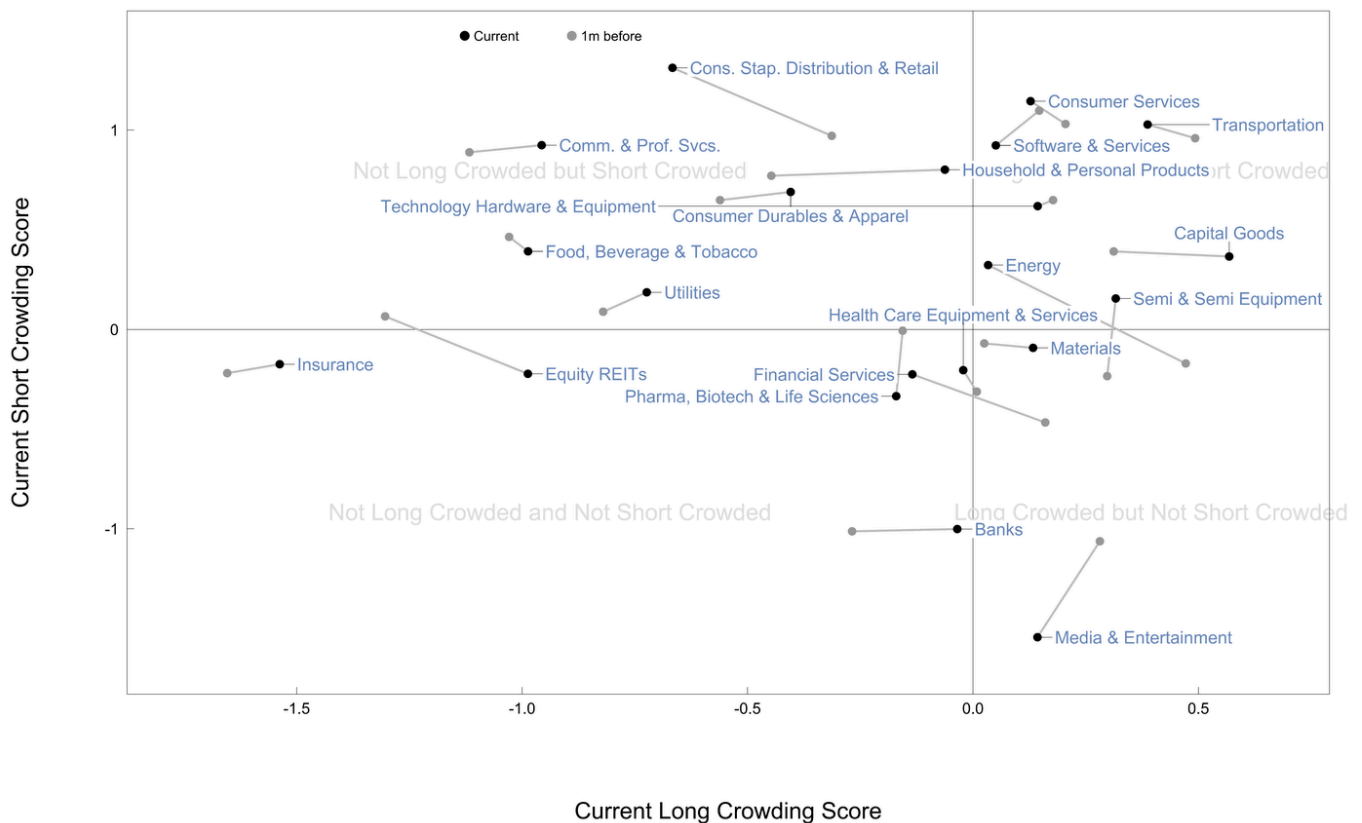


Exhibit 3: Thematic Crowding: Long & Short Side

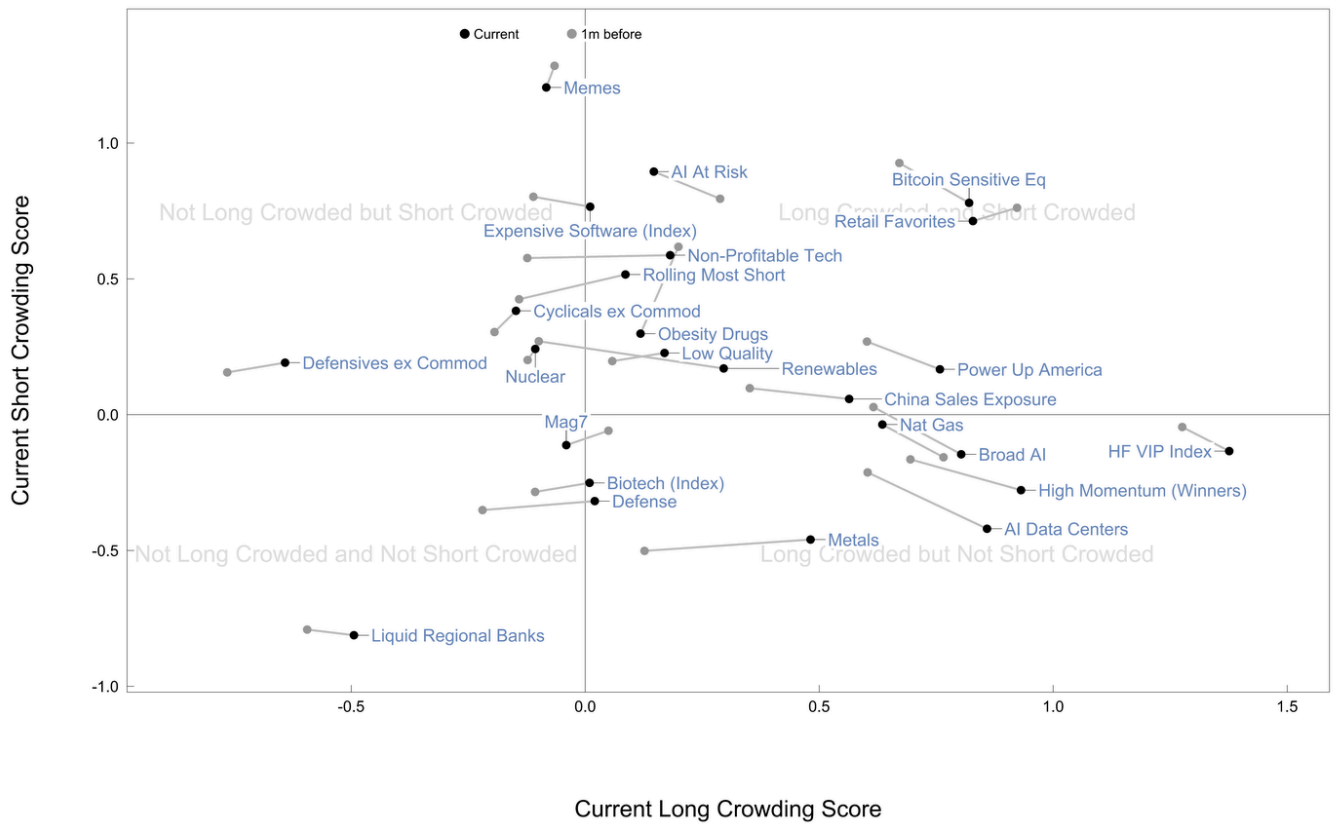


Exhibit 4: Sector Long Crowding: 6-Month Historical Comparison

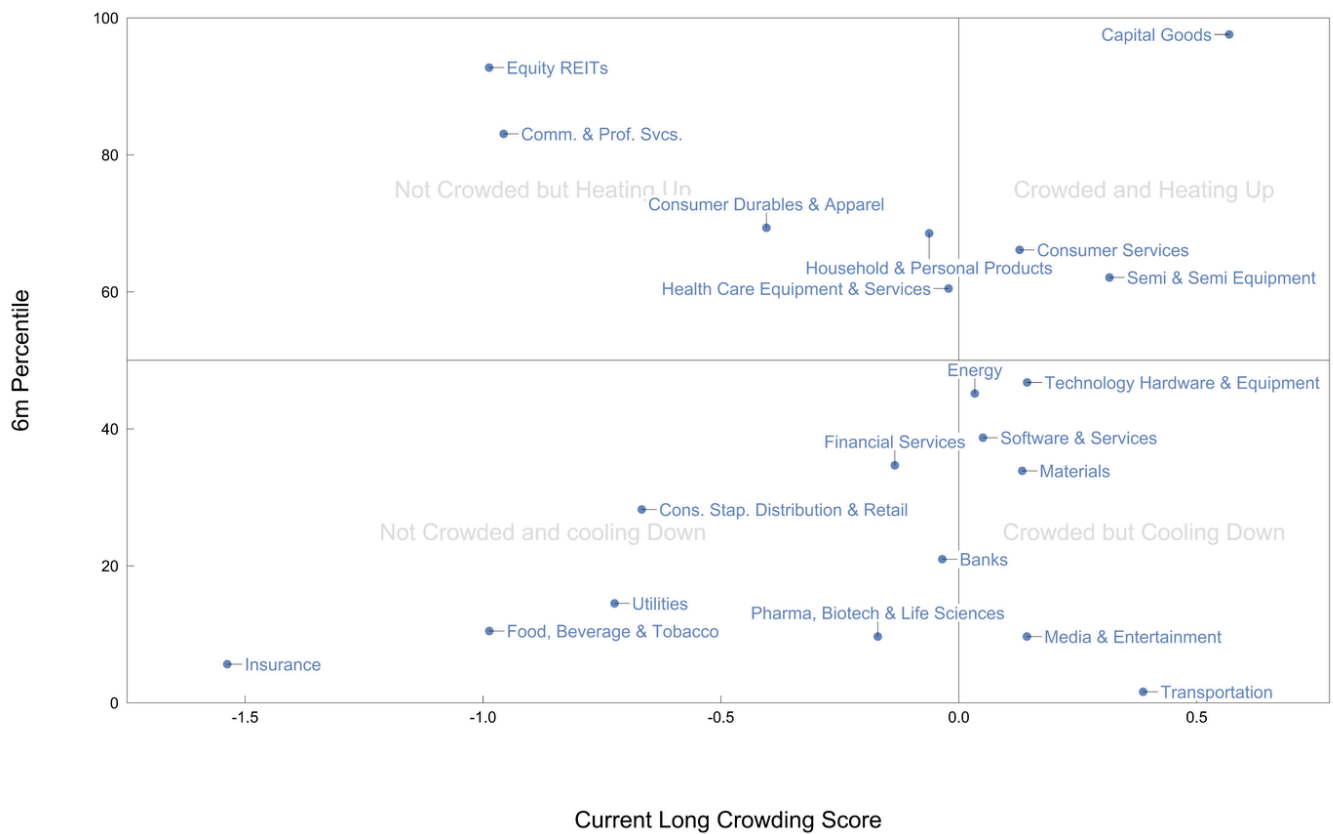


Exhibit 5: Sector Short Crowding: 6-Month Historical Comparison

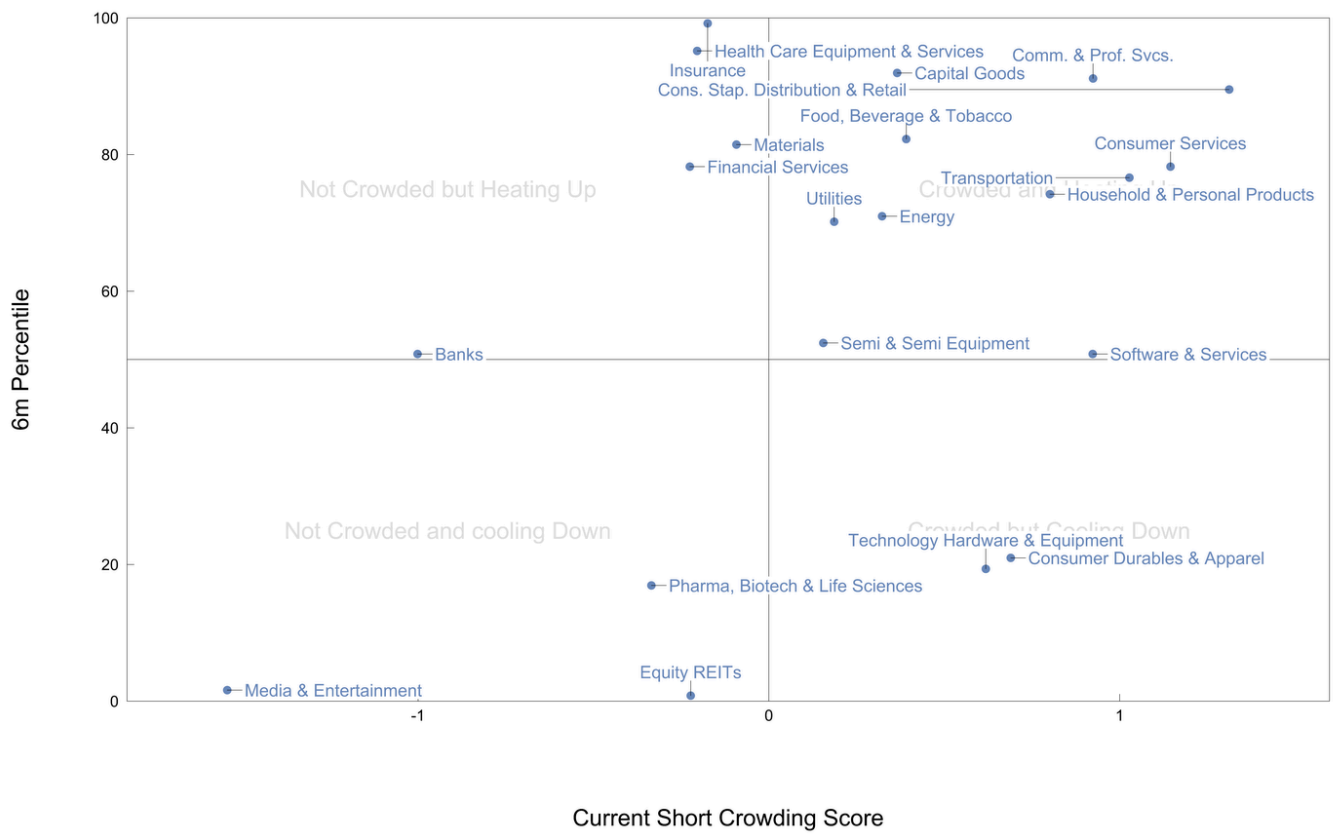


Exhibit 6: Thematic Long Crowding: 6-Month Historical Comparison

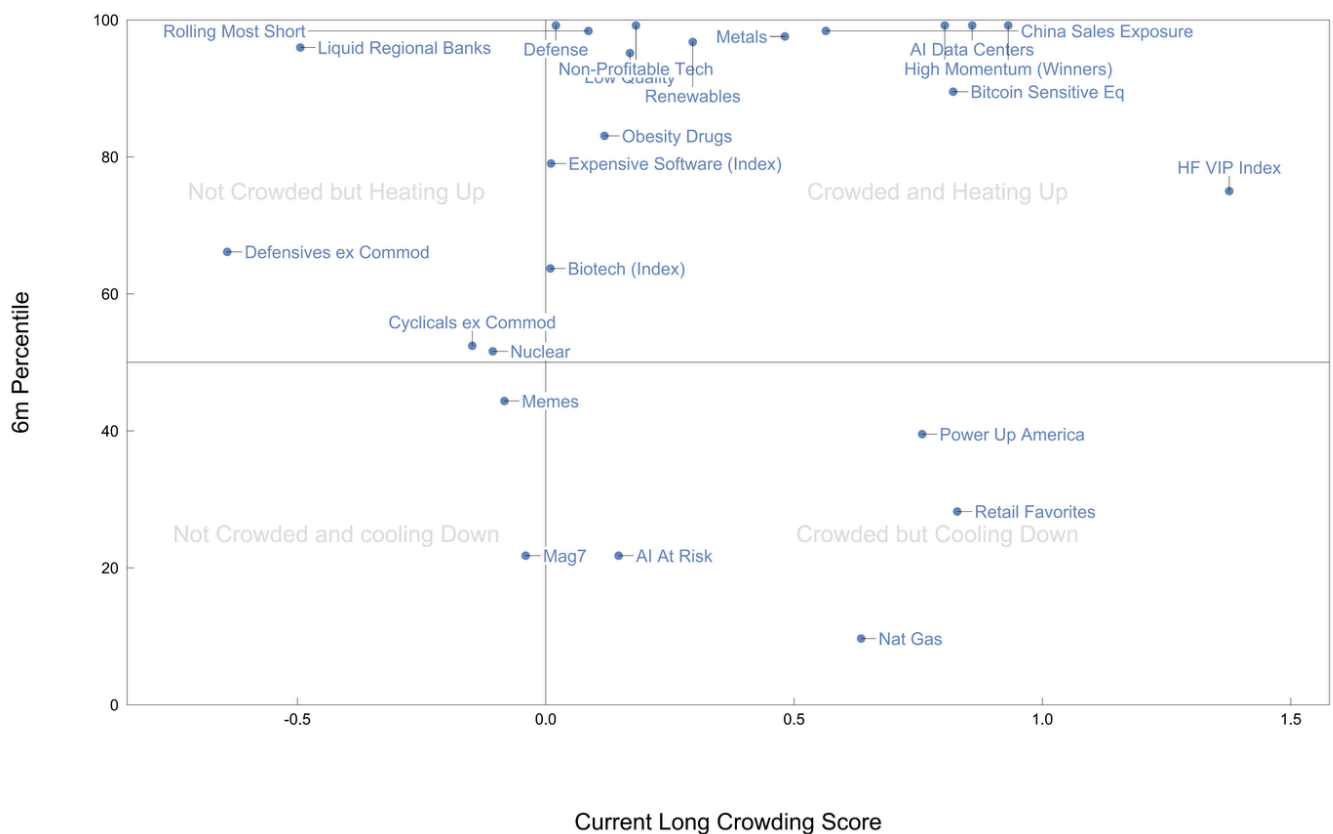
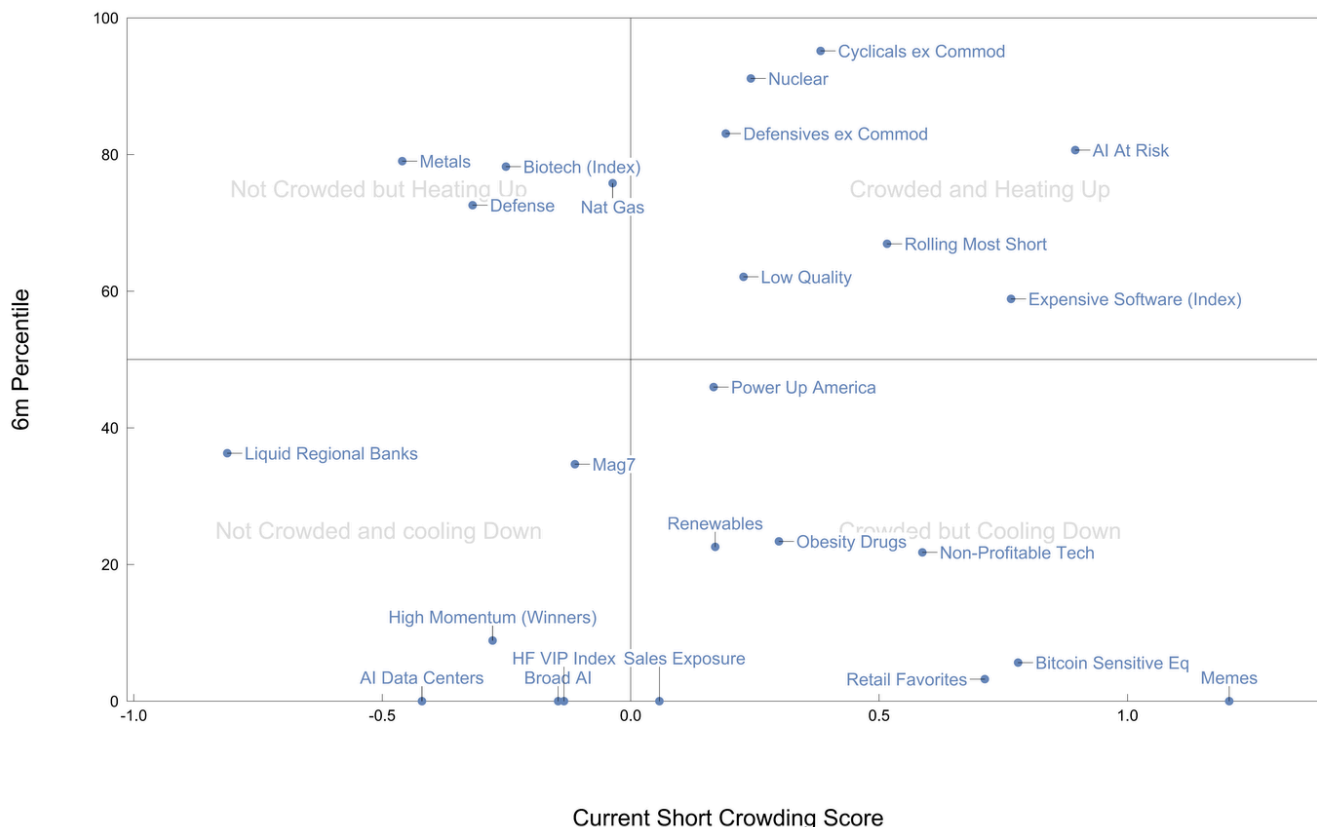


Exhibit 7: Thematic Short Crowding: 6-Month Historical Comparison



Source for all charts: Goldman Sachs FICC and Equities and Prime Services data as of May 21, 2026. Past performance is not indicative of future results.

Methodology and Definitions

Purpose

This report monitors crowding dynamics across US equity sectors and thematic baskets, using aggregated positioning data from Goldman Sachs' prime brokerage platform. It is designed to help identify consensus trades across hedge fund community.

Crowding Score

The crowding score quantifies the degree of hedge fund crowding. It is constructed in three layers:

- Asset-Level Score:** The crowding score is defined separately for the long and short leg of each asset. It measures the number of long (or short) holders in excess of the level expected given the asset's market capitalization.
- Sector and Thematic Aggregation:** The crowding score at the sector or thematic level is calculated as a weighted composite of the underlying asset-level scores.
- Interpretation of Standardized Values:** Asset-level scores are expressed as z-scores — measuring the number of standard deviations from the US stock universe mean — and typically range from -2 to +2. A score above 0 indicates that a sector or theme is crowded, while a score above +1 indicates that it is highly crowded.

Long Crowding vs. Short Crowding

Exhibit 2 and 3 plot crowding scores for sectors and themes, respectively, with long crowding on the x-axis and short crowding on the y-axis. Each exhibit also illustrates the one-month change

in positioning: the darker bubble represents current crowding levels, while the lighter gray bubble represents positioning one month prior.

6-month Historical Comparison

Exhibits 4–7 plot the current crowding score on the x-axis against its percentile rank over the trailing six-month period on the y-axis. This contextualizes whether current positioning is elevated or subdued relative to recent history.

Data Source

Long and short holder data are sourced from Goldman Sachs' prime brokerage book on a daily basis. Thematic exposure is represented using thematic baskets constructed by the GS Equity Custom Baskets Desk.

This message has been prepared by personnel in the Goldman Sachs FICC and Equities business units of one or more affiliates of The Goldman Sachs Group, Inc. ("Goldman Sachs") and is not the product of Global Investment Research. It is not a research report and is not intended as such.

Additional Disclaimers

Non-Reliance and Risk Disclosure: This material is for the general information of our clients and is a solicitation of derivatives business generally, only for the purposes of, and to the extent it would otherwise be subject to, CFTC Regulations 1.71 and 23.605. This material should not be construed as an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. We are not soliciting any specific action based on this material. For the purposes of U.S. rules and regulations, it does not constitute a recommendation or take into account the particular investment objectives, financial conditions, or needs of individual clients. Before acting on this material, you should consider whether it is suitable for your particular circumstances and, if necessary, seek professional advice. The price and value of the investments referred to in this material and the income from them may go down as well as up, and investors may realize losses on any investments. Past performance is not a guide to future performance. Future returns are not guaranteed, and a loss of original capital may occur. We do not provide tax, accounting, or legal advice to our clients, and all investors are advised to consult with their tax, accounting, or legal advisers regarding any potential investment. The material is based on information that we consider reliable, but we do not represent that it is accurate, complete and/or up to date, and it should not be relied on as such. Opinions expressed are our current opinions as of the date appearing on this material only and only represent the views of the author and not those of Goldman Sachs, unless otherwise expressly noted.

Conflict of Interest Disclosure: We are a full-service, integrated corporate advisory, investment management, and brokerage firm. The professionals who prepared this material are paid in part based on the profitability of The Goldman Sachs Group, Inc., which includes earnings from the firm's trading, capital markets, corporate advisory and other business. They, along with other salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients that reflect opinions that are contrary to the opinions expressed herein or the opinions expressed in research reports issued by our Research Departments, and our market making, investing and lending businesses may make investment decisions that are inconsistent with the views expressed herein. In addition, the professionals who prepared this material may also produce material for, and from time to time, may advise or otherwise be part of our trading desks that trade as principal in the securities mentioned in this material. This material is therefore not independent from our interests, which may conflict with your interests. We and our affiliates, officers, directors, and employees, including persons involved in the preparation or issuance of this material, may from time to time have "long" or "short" positions in, act as principal in, and buy or sell the securities or derivatives (including options) thereof in, and act as market maker or specialist in, and serve as a director of, companies mentioned in this material. In addition, we may have served as manager or co manager of a public offering of securities by any such company within the past three years

Not a Fiduciary: To the extent this material is provided to an employee benefit plan or account subject to the Employee Retirement Income Security Act of 1974, as amended ("ERISA") or Section 4975 of the Internal Revenue Code of 1986, as amended (the "Code") (each, a "Benefit Plan Investor"), by accepting this material such Benefit Plan Investor will be deemed to have represented, warranted and acknowledged that (1) the Benefit Plan Investor is represented by a fiduciary that is independent of Goldman Sachs and its affiliates (the "Independent Fiduciary"), (2) the Independent Fiduciary is a bank (as defined in section 202 of the Investment Advisers Act of 1940 (the "Advisers Act")) or a similar institution that is regulated and supervised and subject to periodic examination by a State or Federal agency; an insurance carrier which is qualified under the laws of more than one State to perform the services of managing, acquiring or disposing of assets of a plan; an investment adviser registered under the Advisers Act or, if not registered as an investment adviser under the Advisers Act by reason of paragraph (1) of section 203A, is registered as an investment adviser under the laws of the State in which it maintains its principal office and place of business; a broker-dealer registered under the Securities Exchange Act of 1934; or an Independent Fiduciary that holds or has under management or control total assets of at least \$50 Million; (3) the Independent Fiduciary is capable of evaluating investment risks independently, both in general and with regard to the information contained in this material; (4) the Independent Fiduciary is a fiduciary under ERISA and/or the Code with respect to, and is responsible for exercising independent judgment in evaluating, the information contained in this material; (5) neither Goldman Sachs nor its affiliates are undertaking to provide impartial investment advice, or to give advice in a fiduciary capacity, in connection with the information contained in this material or with respect to any Benefit Plan Investor's subsequent investment decisions; and (6) any financial interest of Goldman Sachs and its affiliates is or has been disclosed. To the extent this material is provided to any other recipient, this material is provided solely on the basis that the recipient has the capability to independently evaluate investment risk and is exercising independent judgment in evaluating investment decisions in that its investment decisions will be based on its own independent assessment of the opportunities and risks presented by a potential investment, market factors and other investment considerations.

Not a Municipal Advisor: Except in circumstances where Goldman Sachs expressly agrees otherwise in writing, Goldman Sachs is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice, including within the meaning of Section 15B of the Securities Exchange Act of 1934.

Phone recording: Telephone conversations with Goldman Sachs personnel may be recorded and retained.

Legal Entities Disseminating this Material: This material is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Canada by either Goldman Sachs Canada Inc. or Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs(Asia) L.L.C. or by Goldman Sachs Asia Bank Limited, a restricted licence bank; in Japan by Goldman Sachs Japan Co.,Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Singapore by Goldman Sachs(Singapore) Pte. (Company Number: 198602165W), by Goldman Sachs Futures Pte.Ltd (Company Number: 199004153Z) or by J.Aron &Company (Singapore) Pte (Company Number:198902119H); in India by Goldman Sachs (India) Securities Private Limited, Mumbai Branch; in the United Kingdom, Belgium, Denmark, Finland, Ireland, Italy, Liechtenstein and Luxembourg by Goldman Sachs International ("GSI"); in the United Kingdom, Belgium, France, Germany, Ireland, Luxembourg and the Netherlands by Goldman Sachs Bank USA, London Branch (Financial Services Register number 582328); in France by Goldman Sachs Paris Inc. et Cie or GSI Paris branch; in Spain by GSI Madrid branch; in Sweden by GSI Stockholm branch; in all European Economic Area jurisdictions by Goldman Sachs Bank Europe SE, including through its authorised branches ("GSBE"); in Switzerland by GSI; in the Cayman Islands by Goldman Sachs (Cayman) Trust, Limited; in Brazil by Goldman Sachs do Brasil Banco Múltiplo S.A.; in South Africa Goldman Sachs International is operating under an exemption in respect of certain clients in terms of the

Financial Advisory and Intermediary Services Act (FAIS), and in the United States of America by Goldman Sachs & Co. LLC (which is a member of FINRA, NYSE and SIPC) and by Goldman Sachs Bank USA. You may obtain information about SIPC, including the SIPC brochure, by contacting SIPC (website: <http://www.sipc.org>; phone: 202-371-8300). GSI, which is authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this material in connection with its distribution in the United Kingdom. GSI, whose registered office is at Plumtree Court, 25 Shoe Lane, London EC4A 4AU, appears in the FCA's Register (Registration No.: 142888). GSI is registered as a Private Unlimited Company in England and Wales (Company Number: 2263951) and its VAT registration number is GB 447 2649 28. GSI is subject to the FCA and PRA rules and guidance, details of which can be found on the websites of the FCA and PRA at www.fca.org.uk and www.bankofengland.co.uk/prd. The FCA is located at 12 Endeavour Square, London E20 1JN, and the PRA is located at Bank of England, 20 Moorgate, London EC2R 6DA. GSBE is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism established between those Member States of the European Union whose official currency is the Euro, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank. Unless governing law permits otherwise, you must contact a Goldman Sachs entity in your home jurisdiction if you want to use our services in effecting a transaction in the securities mentioned in this material. This material is not for distribution to retail clients, as that term is defined under The European Union Markets in Financial Instruments Directive(2014/65/EU) (including as implemented in the United Kingdom), and any investments, including derivatives, mentioned in this material will not be made available by us to any such retail client.

Reproduction and Re-Distribution: Without our prior written consent, no part of this material may be (i) copied, photocopied or duplicated in any form by any means or (ii) redistributed. Notwithstanding anything herein to the contrary, and except as required to enable compliance with applicable securities law, you (and each of your employees, representatives and other agents) may disclose to any and all persons the U.S. federal income and state tax treatment and tax structure of the transaction and all materials of any kind (including tax opinions and other tax analyses) that are provided to you relating to such tax treatment and tax structure, without Goldman Sachs imposing any limitation of any kind.

Information Not for Further Dissemination: To the extent this communication contains Goldman Sachs' pricing information, such pricing information is proprietary and/or confidential and is provided solely for the internal use of the intended recipient(s). You are notified that any unauthorized use, dissemination, distribution or copying of this communication or its contents, including pricing information, in whole or in part, is strictly prohibited. Further, unless prohibited by local law, any use, review or acceptance of this information is subject to and manifests your agreement with Goldman Sachs to use such information only in accordance with the terms set forth above. Goldman Sachs has caused its proprietary information to be delivered to you in reliance upon such agreement.

Not a Valuation: Values herein are not customer valuations and should not be used in lieu of a customer valuation statement or account statement. These values may not reflect the value of the positions carried on the books and records of Goldman Sachs or its affiliates and should not be relied upon for the maintenance of your books and records or for any tax, accounting, legal or other purposes. The information provided herein does not supersede any customer statements, confirmations or other similar notifications.

Receipt of Orders: An order sent to Goldman Sachs by email or instant message is not deemed to be received by Goldman Sachs until a Goldman Sachs representative verifies the order details with a phone call to the client or acknowledges receipt of the order via email or instant message to the client. Goldman Sachs does not accept client orders sent via fax or voicemail systems.

Indicative Terms/Pricing Levels: This material may contain indicative terms only, including but not limited to pricing levels. There is no representation that any transaction can or could have been effected at such terms or prices. Proposed terms and conditions are for discussion purposes only. Finalized terms and conditions are subject to further discussion and negotiation.

Notice to Australian Investors: When this document is disseminated in Australia by Goldman Sachs & Co. LLC ("GSCO"), Goldman Sachs International ("GSI"), Goldman Sachs (Asia) L.L.C. ("GSALLC"), or Goldman Sachs (Singapore) Pte ("GSSP") (collectively the "GS entities"), this document, and any access to it, is intended only for a person that has first satisfied the GS entities that:

- the person is a Sophisticated or Professional Investor for the purposes of section 708 of the Corporations Act of Australia; and
- the person is a wholesale client for the purpose of section 761G of the Corporations Act of Australia.

To the extent that the GS entities are providing a financial service in Australia, the GS entities are each exempt from the requirement to hold an Australian financial services licence for the financial services they provide in Australia. Each of the GS entities are regulated by a foreign regulator under foreign laws which differ from Australian laws, specifically:

- GSCO is regulated by the US Securities and Exchange Commission under US laws;
- GSI is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, under UK laws;
- GSALLC is regulated by the Hong Kong Securities and Futures Commission under Hong Kong laws; and
- GSSP is regulated by the Monetary Authority of Singapore under Singapore laws.

Notice to Brazilian Investors

Marquee is not meant for the general public in Brazil. The services or products provided by or through Marquee, at any time, may not be offered or sold to the general public in Brazil. You have received a password granting access to Marquee exclusively due to your existing relationship with a GS business located in Brazil. The selection and engagement with any of the offered services or products through Marquee, at any time, will be carried out directly by you. Before acting to implement any chosen service or products, provided by or through Marquee you should consider, at your sole discretion, whether it is suitable for your particular circumstances and, if necessary, seek professional advice. Any steps necessary in order to implement the chosen service or product, including but not limited to remittance of funds, shall be carried out at your discretion. Accordingly, such services and products have not been and will not be publicly issued, placed, distributed, offered or negotiated in the Brazilian capital markets and, as a result, they have not been and will not be registered with the Brazilian Securities and Exchange Commission (Comissão de Valores Mobiliários), nor have they been submitted to the foregoing agency for approval. Documents relating to such services or products, as well as the information contained therein, may not be supplied to the general public in Brazil, as the offering of such services or products is not a public offering in Brazil, nor used in connection with any offer for subscription or sale of securities to the general public in Brazil.

The offer of any securities mentioned in this message may not be made to the general public in Brazil. Accordingly, any such securities have not been nor will they be registered with the Brazilian Securities and Exchange Commission (Comissão de Valores Mobiliários) nor has any offer been submitted to the foregoing agency for approval. Documents relating to the offer, as well as the information contained therein, may not be supplied to the public in Brazil, as the offer is not a public offering of securities in Brazil. These terms will apply on every access to Marquee.

Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 380 5764 e/ou contatogoldmanbrasil@gs.com.

Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h. Canal de Denúncias: 0800 047 4065 e/ou Formulário eletrônico: <https://gs.tnwreports.com/>

Public Communication Channel Goldman Sachs Brazil: 0800 380 5764 and/or contatogoldmanbrasil@gs.com.

Available Weekdays (except holidays), from 9am to 6pm. Business Integrity Issues Channel: 0800 047 4065 and/or Web Form: <https://gs.tnwreports.com/>

Note to Investors in Israel: GS is not licensed to provide investment advice or investment management services under Israeli law.

Notice to Investors in Japan

Marquee is made available in Japan by Goldman Sachs Japan Co., Ltd.

本書は情報の提供を目的としております。また、売却・購入が違法となるような法域での有価証券その他の売却若しくは購入を勧めるものでもありません。ゴールドマン・サックスは本書内の取引又はストラクチャーの勧誘を行うものではありません。これらの取引又はストラクチャーは、社内及び法規制等の承認等次第で実際にはご提供できない場合がございます。

<適格機関投資家限定 転売制限>

ゴールドマン・サックス証券株式会社が適格機関投資家のみを相手方として取得申込みの勧誘（取得勧誘）又は売付けの申込み若しくは買付けの申込みの勧誘（売付け勧誘等）を行う有価証券には、適格機関投資家に譲渡する場合以外の譲渡が禁止される旨の制限が付されています。有価証券は金融商品取引法第4条に基づく財務局に対する届出が行われておりません。なお、本告知はお客様によるご同意のもとに、電磁的に交付させていただいております。

<適格機関投資家用資料>

本資料は、適格機関投資家のお客さまのみを対象に作成されたものです。本資料における金融商品は適格機関投資家のお客さまのみがお取引可能であり、適格機関投資家以外のお客さまからのご注文等はお受けできませんので、ご注意ください。商号等/ゴールドマン・サックス証券株式会社 金融商品取引業者 関東財務局長（金商）第69号

加入協会/ 日本証券業協会、一般社団法人金融先物取引業協会、一般社団法人第二種金融商品取引業協会、一般社団法人資産運用業協会

本書又はその添付資料に信用格付が記載されている場合、日本格付研究所（JCR）及び格付投資情報センター（R&I）による格付は、登録信用格付業者による格付（登録格付）です。その他の格付は登録格付である旨の記載がない場合は、無登録格付です。無登録格付を投資判断に利用する前に、「無登録格付に関する説明書」（<http://www.goldmansachs.com/disclaimer/ratings.html>）を十分にお読みください。

If any credit ratings are contained in this material or any attachments, those that have been issued by Japan Credit Rating Agency, Ltd. (JCR) or Rating and Investment Information, Inc. (R&I) are credit ratings that have been issued by a credit rating agency registered in Japan (registered credit ratings). Other credit ratings are unregistered unless denoted as being registered. Before using unregistered credit ratings to make investment decisions, please carefully read "Explanation Regarding Unregistered Credit Ratings" (<http://www.goldmansachs.com/disclaimer/ratings.html>).

Notice to Mexican Investors: Information contained herein is not meant for the general public in Mexico. The services or products provided by or through Goldman Sachs Mexico, Casa de Bolsa, S.A. de C.V. (GS Mexico) may not be offered or sold to the general public in Mexico. You have received information herein exclusively due to your existing relationship with a GS Mexico or any other Goldman Sachs business. The selection and engagement with any of the offered services or products through GS Mexico will be carried out directly by you at your own risk. Before acting to implement any chosen service or product provided by or through GS Mexico you should consider, at your sole discretion, whether it is suitable for your particular circumstances and, if necessary, seek professional advice. Information contained herein related to GS Mexico services or products, as well as any other information, shall not be considered as a product coming from research, nor it contains any recommendation to invest, not to invest, hold or sell any security and may not be supplied to the general public in Mexico.

Notice to New Zealand Investors: When this document is disseminated in New Zealand by Goldman Sachs & Co. LLC ("GSCO"), Goldman Sachs International ("GSI"), Goldman Sachs (Asia) L.L.C. ("GSALLC") or Goldman Sachs (Singapore) Pte ("GSPP") (collectively the "GS entities"), this document, and any access to it, is intended only for a person that has first satisfied; the GS entities that the person is someone:

- (i) who is an investment business within the meaning of clause 37 of Schedule 1 of the Financial Markets Conduct Act 2013 (New Zealand) (the "FMC Act");
- (ii) who meets the investment activity criteria specified in clause 38 of Schedule 1 of the FMC Act;
- (iii) who is large within the meaning of clause 39 of Schedule 1 of the FMC Act; or
- (iv) is a government agency within the meaning of clause 40 of Schedule 1 of the FMC Act.

No offer to acquire the interests is being made to you in this document. Any offer will only be made in circumstances where disclosure is not required under the Financial Markets Conducts Act 2013 or the Financial Markets Conduct Regulations 2014.

Notice to Swiss Investors: This is marketing material for financial instruments or services. The information contained in this material is for general informational purposes only and does not constitute an offer, solicitation, invitation or recommendation to buy or sell any financial instruments or to provide any investment advice or service of any kind.

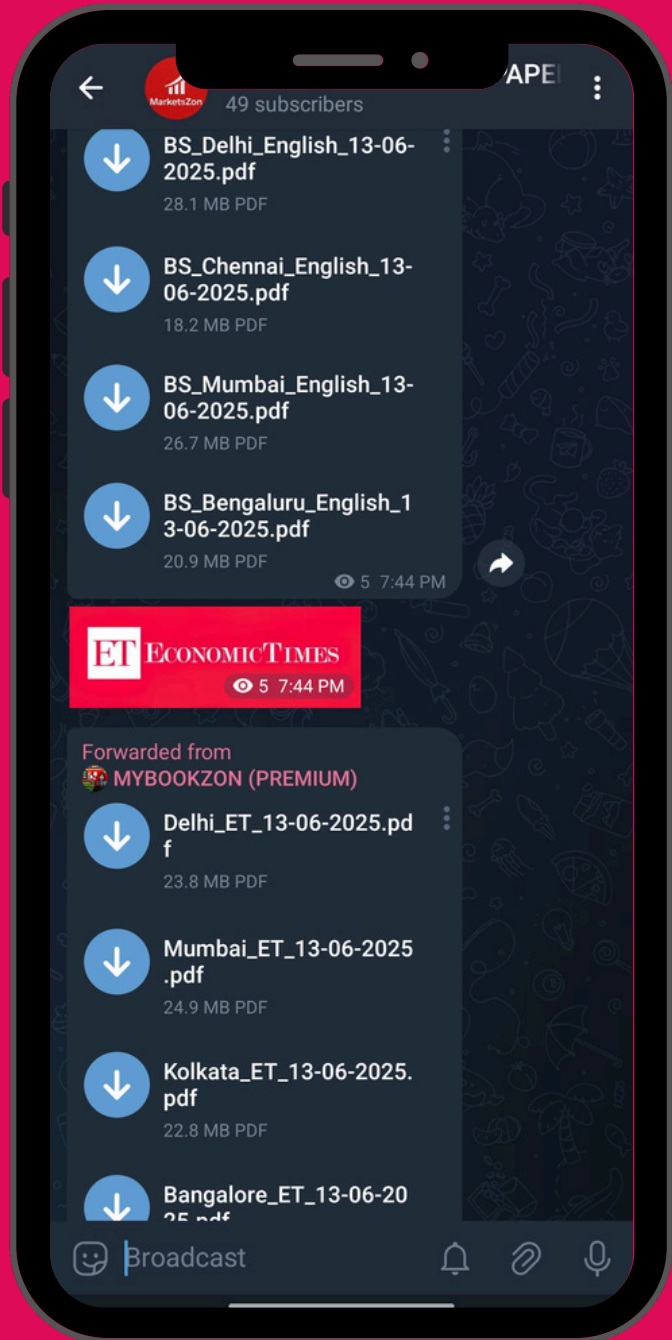
Notice to Investors in Korea: Marquee is made available in Korea by Goldman Sachs (Asia) L.L.C. Seoul Branch.

The information contained in this document does not constitute, and is not intended to constitute, a public offer of securities in the United Arab Emirates in accordance with the Commercial Companies Law (Federal Law no. 2 of 2015), ESCA Board of Directors' Decision no. (9/r.m.) of 2016, ESCA Chairman Decision no 3/r.m. of 2017 concerning promoting and introducing regulations or otherwise under the laws of the United Arab Emirates. Accordingly, the interests in the securities may not be offered to the public in the UAE (including the Dubai International Financial Centre and the Abu Dhabi Global Market). This document has not been approved by, or filed with the Central Bank of the United Arab Emirates, the Securities and Commodities Authority, the Dubai Financial Services Authority, the Financial Services Regulatory Authority or any other relevant licensing authorities in the United Arab Emirates. If you do not understand the contents of this document, you should consult with a financial advisor. This document is provided to the recipient only and should not be provided to or relied on by any other person.

© 2026 Goldman Sachs. All rights reserved.

MarketsZon

FINANCIAL NEWS PAPER, MAGAZINE, RESEARCH REPORTS, COURSE'S, BOOKS, IPO, INVESTMENT REPORTS etc.....



CLICK AND JOIN



ALL FINANCIAL NEWS PAPER'S & MAGAZINE AVAILABLE